26-VAC-201 — Environmental Science

Course Code	26-VAC-201	**Category**	Value-Added Course
Course Title	Environmental Science	**L – T – P**	3 – 0 – 0
Credits	3	**Contact Hours**	42
Semester	Second	**Assessment**	Theory: CA 30 + ESE 70 (with a compulsory field / activity portfolio in the WCA)

Prerequisites

A compulsory course for every undergraduate under the direction of the Hon'ble Supreme Court of India and the consequent UGC syllabus, here integrated with the University Grants Commission's draft Guidelines and Curriculum Framework on Mission LiFE (Lifestyle for Environment). It is self-contained. Assumed:

- The general science, geography and civics of Class X (WBBSE / CBSE / ICSE).
- No Higher Secondary environmental science is assumed.
- A smartphone or shared institutional device for the Meri LiFE portal and app activities of Module V; the institution arranges access for any student who lacks one.

Course Objectives

- To establish the structure and functioning of natural systems and the environmental crisis the engineer's work must reckon with.
- To ground the student in India's environmental law, policy, national missions and the UN Sustainable Development Goals.
- To introduce Mission LiFE — its genesis, seven themes, and the theory and practice of behavioural change.
- To convert the above into measured practice through footprint accounting, environmental audits and a documented behavioural-change portfolio.

Course Outcomes

On completion of the course the student will be able to:

CO	Statement	Bloom
CO1	Explain the structure and functioning of ecosystems, the value of biodiversity and the ecosystem services on which human activity depends.	L2 – Understand
CO2	Analyse the principal forms of environmental pollution and the triple planetary crisis and relate them to lifestyle and consumption.	L4 – Analyse
CO3	Explain India's environmental law, policy, national missions and SDG commitments.	L2 – Understand
CO4	Explain Mission LiFE — its genesis, pillars of Demand, Supply and Policy, and seven themes — and the principal theories of behavioural change.	L3 – Apply
CO5	Conduct a footprint calculation and an environmental audit and design and document a behavioural-change intervention using the Meri LiFE framework.	L5 – Evaluate

CO–PO Articulation Matrix

Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	1	1	–	3	1	–	1	–	2
CO2	3	3	1	2	1	3	1	–	1	–	2
CO3	2	2	1	1	–	3	2	–	1	1	2
CO4	2	2	2	1	1	3	3	1	2	1	3
CO5	2	3	3	3	2	3	2	2	2	1	3

Course Plan — Module-wise Syllabus

Module	Title	Contents	Hours

Module	Title	Contents	Hours
I	The Environment, Ecosystems and Biodiversity	The multidisciplinary nature of environmental studies; natural resources — forest, water, soil, mineral, food and energy — and the consequences of their over-use. The structure and function of an ecosystem; energy flow, the food chain and the ecological pyramid. Biodiversity — genetic, species and ecosystem; its value and the ecosystem services it renders; threats and conservation, with reference to India as a mega-diverse country.	9
II	Environmental Pollution and the Global Challenges	Air, water, soil, noise and thermal pollution — sources, effects and control; solid, hazardous and electronic waste and its management; the triple planetary crisis of climate change, pollution and biodiversity loss; global water resources, microplastics and groundwater over-extraction; the linkage between lifestyle and consumption and environmental impact, and the ecological and carbon footprint. [Mission LiFE Units I–II.]	8
III	India's Sustainability Policy and Legal Framework	The constitutional provisions and the principal environmental legislation — the Water, Air and Environment (Protection) Acts, the Wildlife Protection and Forest Conservation Acts — and the institutional framework; India's climate policy, Nationally Determined Contributions and SDG commitment. India's national missions — the National Solar Mission, Green Hydrogen Mission, Jal Jeevan Mission, Atal Bhujal Yojana, Smart Cities Mission and AMRUT — and their relation to Mission LiFE. [Mission LiFE Unit III.]	8
IV	Mission LiFE and Behavioural Change	Mission LiFE (Lifestyle for Environment) — context, genesis, launch, and promotion nationally and internationally; the pillars of Demand, Supply and Policy; sustainable living as a mass movement and the circular economy of mindful consumption. The role of individual and collective behaviour; theories of behavioural change and their maintenance; the design of a behavioural nudge; India's own traditions of sustainability. [Mission LiFE Units IV and, in part, I.]	8
V	The Seven Themes of Mission LiFE, and Practice	The seven themes — Save Energy, Save Water, Reduce Single-Use Plastic, Adopt Sustainable Food Systems, Reduce Waste, Adopt Healthy Lifestyles, and Reduce E-Waste — and the eco-friendly practices under each. Communication for behavioural change; the Meri LiFE portal and app and other innovations for mass engagement; community engagement and social responsibility. The whole-semester component of the course is the field-and-activity portfolio built on this module. [Mission LiFE Units V–VI.]	9

Cross-cutting notes. This course delivers the University's environmental-education obligation and the national Mission LiFE programme together; the bracketed references map the modules to the six Mission LiFE units, so a college satisfies both from a single course. The field-and-activity portfolio is compulsory: a student who does not submit it is not to be passed, since both requirements are directed at practice. It connects to the Environment-and-Sustainability graduate attribute (PO6, The Engineer and the World) and to the water, effluent and cleaner-production courses of the later years.

Text and Reference Books

- M. P. Poonia, Environmental Studies, Khanna Book Publishing Co. / AICTE Recommended Textbook.
- Erach Bharucha, Textbook of Environmental Studies for Undergraduate Courses, UGC / Universities Press.
- University Grants Commission, Guidelines and Curriculum Framework on Mission LiFE at the Undergraduate Level (draft) — integrated into Modules II–V.
- Subrat Roy, Environmental Science, Khanna Publishing House (AICTE Prescribed Textbook).
- NITI Aayog, LiFE — Lifestyle for Environment; and Thinking for Our Planet: 75 Ideas to Promote LiFE.
- MoEFCC, Parampara — India's Culture of Climate-Friendly Sustainable Practices; and Prayaas Se Prabhaav Tak (MoEFCC and UNDP, 2022).
- IPCC, Synthesis Report of the Sixth Assessment Report (2023); India State of Forest Report, Forest Survey of India.
- Y. Anjaneyulu, Introduction to Environmental Science, BS Publications.

Suggested SWAYAM / NPTEL and Online Resources

- Mission LiFE portal and Meri LiFE app (missionlife-moefcc.nic.in; merilife.nic.in) — for the activity and reporting component.
- SWAYAM / NPTEL — Environmental Studies / Sustainability, as offered.
- WWF and Global Footprint Network carbon- and ecological-footprint calculators — for the footprint activities.

Question-paper note. The end-semester paper covers Modules I–IV; Module V is assessed through the portfolio, not by written examination, since it is a module of practice. At least one Group C question relates a lifestyle or consumption pattern to its environmental consequence and to a Mission LiFE theme. Prescribed data — emission and footprint factors — are printed with the paper.